

SAT.O–Blockwave — Native Covariant Equations

Full 4D geometric rewrite: unified variational construct, EL equations, and mapping log

This document gives a single, fully 4D-covariant presentation of the SAT/Blockwave (SAT_O) framework. It (i) defines the emergent spacetime geometry from filamentary/twist fields, (ii) writes a single unified action built from purely geometric and topological SAT fields, (iii) lists the Euler–Lagrange equations (Einstein-like, gauge, Dirac, scalar, and foliation/strain), (iv) provides explicit identifications (mapping log) between standard physics structures and SAT constructs, and (v) flags key deviations and operational interpretations.

A. Axioms and primitives (SAT-native)

Work on a 4-dimensional differentiable manifold M (no prior metric). The basic SAT primitives are:

1. A *filament bundle frame field* (local 1-form/tetrad sections) $E^I_\mu(x)$, $I = 0, \dots, 3$. These are fields on M that encode local filament directions and packing. They are not assumed orthonormal a priori; an internal Minkowski form $\eta_{IJ} = \text{diag}(-1, 1, 1, 1)$ is introduced as an algebraic fiber metric to build a spacetime metric from the frame:

$$g_{\mu\nu}(x) \equiv \eta_{IJ} E^I_\mu(x) E^J_\nu(x).$$

The tetrad is *emergent* (a functional of filament microstate), but treated here as a field variable for the variational principle.

2. A *unit time-flow vector* (foliation) $u^\mu(x)$. In SAT this is identified with the timelike frame direction:

$$u^\mu = \frac{E^{0\mu}}{\sqrt{-g_{\alpha\beta} E^{0\alpha} E^{0\beta}}}, \quad u^\mu u_\mu = -1.$$

(Enforced by a multiplier in the action.)

3. A compact *phase field* $\psi(x) \in S^1$ (U(1)-valued) and a misalignment angle $\theta_4(x) \in S^1$ controlling disformal factors and phase-locking.
4. A discrete *twist/topological field* $\tau(x) \in \mathbb{Z}_N$ (locally encoded as integer-valued defects or as a density current of linking).
5. A *linking density* $\rho_G(x)$ and derived topological invariant $Q(x)$ built from filament linking/writhe/winding; these appear as dimensionless scalar densities.
6. Gauge connection 1-forms $A_\mu^A(x)$ (with A an index in the emergent Lie algebra, e.g. $\mathfrak{u}(1), \mathfrak{su}(N)$), and fermionic spinor fields $\Psi(x)$ defined on the spinor bundle induced by the tetrad.

All tensor operations (raising/lowering, covariant derivatives) use the Levi-Civita connection ∇_μ built from $g_{\mu\nu}$ (which itself is built from E^I_μ). Thus there is no background metric independent of filament/tetrad fields.

B. The unified SAT action

Define the single action (all fields dynamical):

$$S_{\text{SAT}}[E, u, \psi, \theta_4, \tau, A, \Psi, \Phi] = \int_M d^4x \sqrt{-g} \left[\frac{c^4}{16\pi G_{\text{ind}}} R[g] + \mathcal{L}_u + \mathcal{L}_{\theta_4} + \mathcal{L}_\psi + \mathcal{L}_{\text{gauge}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{int}} \right] + S_{\text{Top}}.$$

All pieces are SAT-native: G_{ind} will be derived (parametrically) from filament microparameters; S_{Top} collects discrete topological contributions (proportional to $\hbar$). We now define each block in manifestly covariant form (indices raised/lowered with g).

B.1 Geometry / tetrad sector

Tetrad determinant $e = \det(E^I_\mu)$, metric $g_{\mu\nu} = \eta_{IJ} E^I_\mu E^J_\nu$. The Ricci scalar built from g is $R[g]$. We include an explicit tetrad-regularization term (to penalize filament shear at microscale):

$$\mathcal{L}_{\text{tetrad}} = -\frac{\zeta}{2\ell_f^2} g^{\mu\alpha} g^{\nu\beta} (\nabla_\mu E^I_\nu) (\nabla_\alpha E_{I\beta}),$$

where ζ is dimensionless and ℓ_f is the microscopic filament length.

B.2 Foliation / strain block $\mathcal{L}_u$

Define the strain (timesheet) tensor

$$S_{\mu\nu} = \nabla_\mu u_\nu.$$

Use the minimal derivative action:

$$\mathcal{L}_u = -\frac{\kappa}{\ell_f^2} \left(S_{\mu\nu} S^{\mu\nu} - \lambda (S^\mu_\mu)^2 \right) + \Lambda_0 + \Lambda_\lambda (u^\mu u_\mu + 1).$$

Λ_λ enforces $u^\mu u_\mu = -1$. All contractions use g .

B.3 Compact-phase and misalignment $\mathcal{L}_\psi, \mathcal{L}_{\theta_4}$

$$\mathcal{L}_\psi = -\frac{f_\psi^2}{2c^2} (\nabla_\mu \psi) (\nabla^\mu \psi) - V_\psi(\psi), \quad \mathcal{L}_{\theta_4} = -\frac{f_\theta^2}{2c^2} (\nabla_\mu \theta_4) (\nabla^\mu \theta_4) - V_\theta(\theta_4).$$

B.4 Gauge block $\mathcal{L}_{\text{gauge}}$

Let $F_{\mu\nu}^A = \partial_\mu A_\nu^A - \partial_\nu A_\mu^A + f^{ABC} A_\mu^B A_\nu^C$ (covariantized with ∇ where necessary). The effective inverse coupling is a local function of linking density:

$$g_{\text{eff}}^{-2}(x) = \mathcal{B} \rho_G(x) \ell_f^2, \quad e_{\text{eff}}^{-2}(x) = \mathcal{B}' \rho_G(x) \ell_f^2.$$

Then

$$\mathcal{L}_{\text{gauge}} = -\frac{1}{4} g_{\text{eff}}^{-2}(x) F_{\mu\nu}^A F^{A\mu\nu} - \frac{1}{4} e_{\text{eff}}^{-2}(x) F_{\mu\nu} F^{\mu\nu}.$$

B.5 Dirac sector

Spin connection ω_μ^{IJ} determined by tetrad. Covariant derivative on spinors $D_\mu = \partial_\mu + \frac{1}{4}\omega_\mu^{IJ}\gamma_{IJ} + ig_{\text{eff}}A_\mu^AT^A + ie_{\text{eff}}A_\mu$. The Dirac Lagrangian:

$$\mathcal{L}_{\text{Dirac}} = \bar{\Psi} \left(i\hbar \gamma^\mu D_\mu - m_\Psi(x)c \right) \Psi,$$

with emergent mass

$$m_\Psi(x) = \frac{T \ell_f}{c^2} \frac{1}{Q(x)}.$$

Here T is filament tension; $Q(x)$ is topological invariant built from linking/writhe.

B.6 Topological action S_{Top}

Topological couplings (axion-like and linking terms):

$$S_{\text{Top}} = \hbar \int_M \sqrt{-g} \left[\gamma_1 \psi \frac{1}{8\pi^2} F^A \wedge F^A + \gamma_2 \tau \mathcal{I}_{\text{link}} \right].$$

In components the first term is $\propto \hbar \int \sqrt{-g} \psi F_{\mu\nu}^A \tilde{F}^{A\mu\nu} d^4x$.

B.7 Interaction block $\mathcal{L}_{\text{int}}$

Contains Yukawa-like couplings, couplings between foliation strain and other fields, and topological mass operators:

$$\mathcal{L}_{\text{int}} = -y(\psi, \theta_4, \tau) \bar{\Psi} \Psi \Phi_{\text{top}}(\theta_4, \tau) + \chi_1 S^{\mu\nu} T_{\mu\nu}^{(\text{gauge})} + \dots$$

with y determined by topology; Φ_{top} is scalar built from linking invariants.

C. Euler–Lagrange equations (SAT-native covariant field equations)

Vary S_{SAT} w.r.t. the independent fields $E^I_\mu, u^\mu, \psi, \theta_4, A_\mu^A, \bar{\Psi}, \Psi$ (and Lagrange multipliers). Present major resulting equations.

C.1 Emergent Einstein equations (metric variation / tetrad variation)

Varying tetrad (or equivalently metric) yields:

$$\frac{c^4}{8\pi G_{\text{ind}}} G_{\mu\nu}[g] = T_{\mu\nu}^{(u)} + T_{\mu\nu}^{(\theta)} + T_{\mu\nu}^{(\psi)} + T_{\mu\nu}^{(\text{gauge})} + T_{\mu\nu}^{(\text{Dirac})} + T_{\mu\nu}^{(\text{int})} + T_{\mu\nu}^{(\text{tetrad})},$$

where each $T_{\mu\nu}^{(\cdot)}$ is the stress-energy tensor derived in the usual covariant way: $T_{\mu\nu} = -(2/\sqrt{-g})\delta(\sqrt{-g}\mathcal{L})/\delta g^{\mu\nu}$. The left-hand prefactor uses an induced G_{ind} determined by strain/tetrad microphysics (see Sec. E).

C.2 Foliation/strain EOM (variation w.r.t. u^μ)

Variation of $\mathcal{L}_u$ (including coupling to other sectors) yields the nonlinear elastic-timeflow equation:

$$\boxed{\frac{\kappa}{\ell_f^2} \left(\nabla^\alpha S_{\alpha\mu} - \lambda \nabla_\mu S^\alpha{}_\alpha \right) + \mathcal{C}_\mu[u, \psi, \theta_4, A, \Psi] + 2\Lambda_\lambda u_\mu = 0,}$$

where $\mathcal{C}_\mu$ collects coupling currents (exchange of momentum with gauge, scalar, and Dirac sectors). Linearizing around a background $u^\mu = \bar{u}^\mu + \delta u^\mu$ gives wave operators whose speeds depend on κ/ℓ_f^2 and background coefficients.

C.3 Scalar / phase EOMs

$$\boxed{f_\psi^2 \square_g \psi + \frac{\hbar}{8\pi^2} F_{\mu\nu}^A \tilde{F}^{A\mu\nu} + \frac{\partial V_\psi}{\partial \psi} + \mathcal{J}_\psi = 0,}$$

$$\boxed{f_\theta^2 \square_g \theta_4 + \frac{\partial V_\theta}{\partial \theta_4} + \mathcal{J}_\theta = 0,}$$

where $\square_g \equiv g^{\mu\nu} \nabla_\mu \nabla_\nu$ and $\mathcal{J}$ are source terms from interactions.

C.4 Yang–Mills / Maxwell equations (gauge variation)

$$\boxed{\nabla_\mu (g_{\text{eff}}^{-2}(x) F^{A\mu\nu}) + f^{ABC} g_{\text{eff}}^{-2}(x) A_\mu^B F^{C\mu\nu} = J_{\text{matter}}^{A\nu} + \frac{\hbar \gamma_1}{8\pi^2} \nabla_\mu (\psi \tilde{F}^{A\mu\nu}),}$$

and the abelian Maxwell form

$$\boxed{\nabla_\mu (e_{\text{eff}}^{-2}(x) F^{\mu\nu}) = J_{\text{EM}}^\nu + \frac{\hbar \gamma_1}{8\pi^2} \nabla_\mu (\psi \tilde{F}^{\mu\nu}).}$$

Note the explicit dependence of effective couplings on $\rho_G(x)$ makes these equations spatially varying in coupling — a SAT-native hallmark.

C.5 Dirac equation (fermion variation)

$$\boxed{(i\hbar \gamma^\mu D_\mu - m_\Psi(x) c) \Psi = 0,}$$

with covariant derivative D_μ including the spin connection and emergent gauge couplings. The mass $m_\Psi(x) = T\ell_f/(c^2 Q(x))$ depends on topological invariant $Q(x)$.

C.6 Coupled-mode kernel (linearization compact form)

Collect small perturbations $\Psi \equiv (\delta E, \delta u, \delta \psi, \delta \theta_4, \delta A, \delta \Psi)$. Symbolically:

$$\boxed{(\square_g \mathbb{I} + \mathcal{K}[g, u, \psi, \theta_4, \rho_G, \dots]) \Psi = 0,}$$

where $\mathcal{K}$ is a matrix differential operator encoding mode couplings (strain–metric mixing, axion–gauge mixing, Yukawa masses, etc.). The dispersion relations follow from $\det(-\omega^2 + |\mathbf{k}|^2 + \tilde{\mathcal{K}}(\omega, \mathbf{k})) = 0$.

D. SAT $\rightarrow$ Standard-Model mapping log (short)

Below are principal identifications (import log) used to interpret SAT equations operationally.

- **Spacetime metric** $g_{\mu\nu} \longleftrightarrow$ emergent bilinear of filament tetrads E^I_μ .
- **Levi-Civita connection** $\nabla \longleftrightarrow$ metric connection computed from $g_{\mu\nu}$ (no background).
- **Gravitational degrees of freedom** $\longleftrightarrow$ long-wavelength modes of the strain tensor $S_{\mu\nu}$ and slow variations in E^I_μ .
- **Electromagnetic field strength** $F_{\mu\nu} \longleftrightarrow$ antisymmetric part of twist/shear in filament congruences (projected 2-form density built from filament linking currents).
- **Yang–Mills gauge group** $\longleftrightarrow$ algebra of filament linking classes (e.g. $\mathbb{Z}_3$ filament linking $\rightarrow$ SU(3) color via an emergent Lie algebra embedding).
- **Charge quantization** / $e, \hbar \longleftrightarrow$ compactness of ψ , topological coupling coefficients in S_{Top} .
- **Particle mass** $m \longleftrightarrow$ topological suppression $m \sim T\ell_f/(c^2Q)$ with Q the linking/writhe/winding invariant.
- **Axion/CP terms** $\longleftrightarrow \psi F\tilde{F}$ coupling arising naturally from topological coupling of compact phase ψ to gauge instanton density.

E. Emergent constants (parametric relations)

These are scaling relations (symbolic); numeric values require microphysical calculation.

$$\boxed{G_{\text{ind}} \sim \frac{c^4 \ell_f^2}{16\pi \mathcal{A} \kappa}, \quad g_{\text{eff}}^{-2} \sim \mathcal{B} \rho_G \ell_f^2, \quad m \sim \frac{T \ell_f}{c^2 Q}.$$

Here $\mathcal{A}, \mathcal{B}$ are coarse-graining geometry factors computed from filament ensemble averages. $T, \ell_f, \kappa, \rho_G$ are microscopic physical parameters (tension, filament length, strain stiffness, linking density).

F. Operational interpretation and observables

- **Metric observables:** geodesics, curvature invariants $R, R_{\mu\nu}R^{\mu\nu}$, gravitational waves correspond to collective strain excitations $\delta u, \delta E$.
- **Gauge observables:** field strengths $F_{\mu\nu}$, flux integrals, linking densities measurable as Aharonov–Bohm phases (via ψ couplings).
- **Particle masses / spectra:** computed from Q classification of minimal topological excitations and their local tension $\times$ length scales.
- **Coupling-space variability:** spatial/temporal variation in $\rho_G(x)$ yields running and local modulation of effective gauge coupling constants.

G. Deviation flags and caveats

1. **No strict Lorentz background:** Lorentz invariance is emergent; tests of Lorentz symmetry bound the allowed anisotropy and gradient scale of u^μ , E^I_μ .
2. **Coupling locality:** $g_{\text{eff}}^{-2}(x)$ depending on $\rho_G(x)$ makes gauge equations non-uniform; renormalization-group treatment must incorporate topological-statistical inputs.
3. **Spin structure:** spin-bundle existence requires tetrad to be globally orientable and admit spin structure; topologically nontrivial filament configurations may obstruct a global spin structure — physical interpretation: fermions live in regions where spin-bundle trivialization holds (defects = exotic states).
4. **Numerical prefactors:** derivations above give parametric scaling. To obtain precise $G, \hbar, e, m$ one must compute coarse-graining integrals over the filament microstate ensemble (partition-function calculation).
5. **Stability/ghosts:** ensure positivity of kinetic terms (no ghosts) by constraining $\kappa, f_\psi, f_\theta > 0$ and verifying sign of induced Einstein term.

H. Recipe to operationalize (brief)

1. Choose a filament micromodel (loop ensemble with tension T , bending rigidity, linking weights).
2. Compute ensemble averages: $\mathcal{A}, \mathcal{B}, \rho_G(x)$ and coarse-grained stress-energy contributions.
3. Insert into the SAT action; compute $G_{\text{ind}}, g_{\text{eff}}(x)$, mass spectrum $m(Q)$.
4. Linearize around appropriate background and obtain mode spectra; compare with observed GR/SM data to bound ℓ_f, κ, T .

Closing note. The above is a compact, fully covariant SAT-native rewrite of gravity, gauge, scalar and fermion dynamics. It keeps strictly to 4D tensor form, uses only fields emergent from filament/tetrad/topology primitives, and presents explicit equations ready for a micro-to-macro coarse-graining program. Numerical realization requires specification of a filament statistical microphysics and computation of the coarse-graining kernels $\mathcal{A}, \mathcal{B}, \rho_G$.